Meaning in Geometry: Words as Trajectories in Phase Space

The Train of Thought

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From words to metaphors, in this article we travel from sounds to trains of thoughts. And on these trains of thought we find we are on a journey where the curves of the landscape are not only language, but also translation, mathematics, chaos, and even hallucinations.

How Can Space can Carry Meaning?

It seems that hen we speak or write, it feels like we’re passing around packets of meaning, that is discrete words and sentences. It’s as if they were coins we could simply hand over. But when we look close through the eyes of mathematics we find meaning is not stored inside words like money in a purse, meaning can be shown to lives in the geometry of relationships between them.

A powerful way to understand this concept is through **phase space embedding**, as described by ***Takens' theorem***. Phase space is a mathematical space in which each point represents a complete state of a complex system: like language. That last sentence is a complex idea, and it might not quite slot into place like a coin in a purse. This is why why we may need to build up the context, the surrounding coins, so you can hold on to the train of thought.

But what does that idea really mean? What is Takens' Theorem saying?

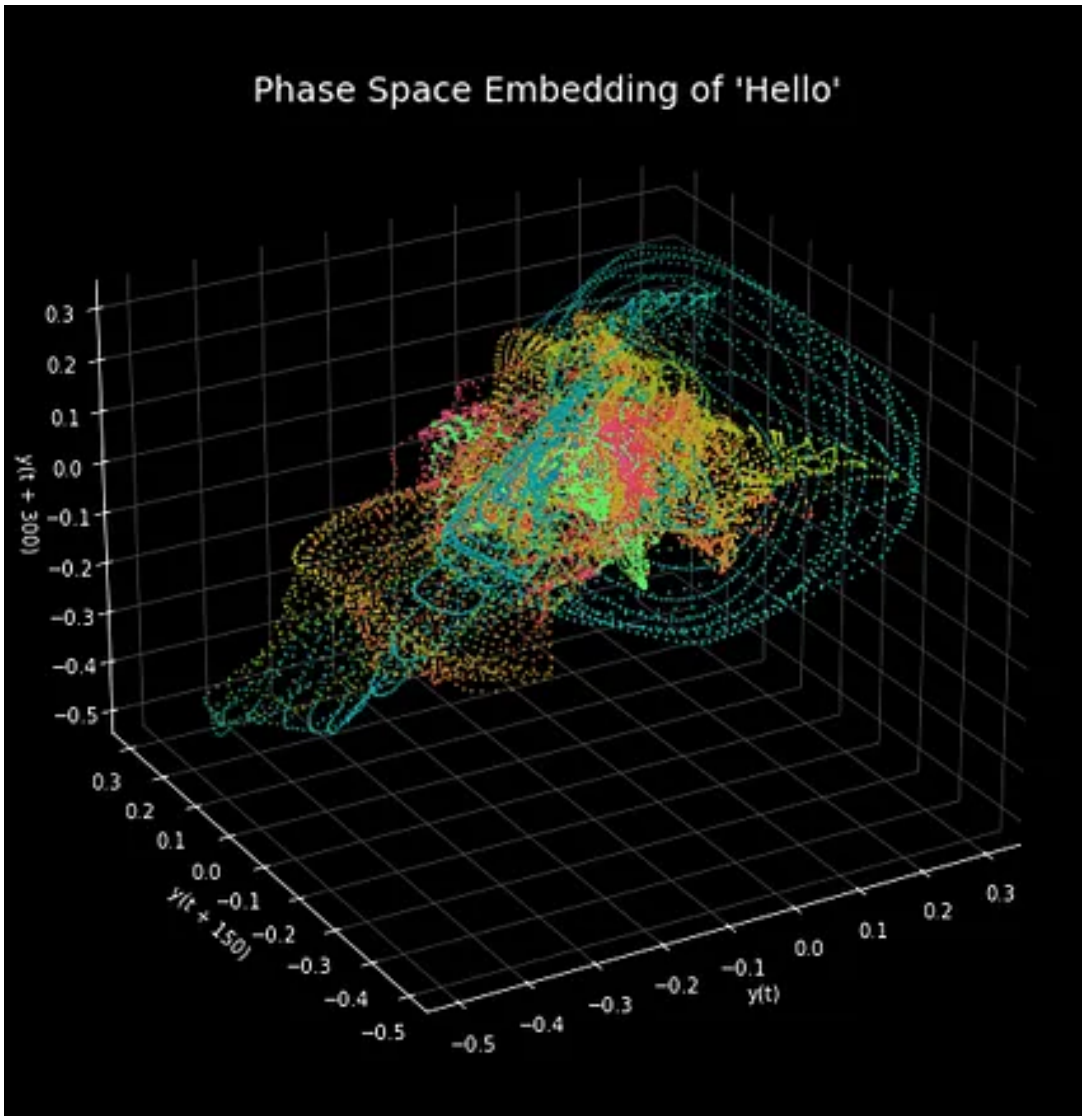
It means that even if we can only measure one variable such as the pressure in an engine, or the amplitude of a sound in a spoken word, Takens showed that we can reconstruct the system's full geometry by plotting delayed versions of that variable against itself.

So in essence we are saying that from a one-dimensional **time series**, a sequential set of measurements, we can rebuild the shape of the **dynamics** of the complex system that produced it. That is to say how the system works. It's an amazing mathematical theorem that was developed by Taken in 1981, when **Chaos Theory** was all the rage.

From sound to geometry

Let's consider the recorded sound of the word *"hello"*. The raw recording is a one-dimensional sequence: air pressure values over time. Applying Takens' method of delays, we sample the recording by taking measurement every few milliseconds, and then we plot each sample against its delayed copies. Then if we use the right delay and embedding dimension, a structure appears in three-dimensional space. We call this structure a geometric "attractor" and this is the signature of that sound.

Importantly, the shape of the system is not arbitrary. It encodes the relationships between the frequencies, timing, and amplitudes that make the sound recognisable as *"hello"*. Change the intonation or accent, and the geometry shifts subtly.



From words to trajectories

Now, we can move on, if we treat each spoken word as a measurement in a larger linguistic system, the same principle applies. A single word is like a small geometric knot in phase space. However, a sequence of words in a sentence becomes a **trajectory** through that space. And here the magic begins as the geometry of that trajectory encodes the relationships between words in context, much as the geometry of *"hello"* encodes the relationships between frequencies in the sound.

In practice we don't talk using single words, we turn the sounds of the sentence into neurological signals using our ears. Our ears transduce the linear stream of words into this geometry. Then we compare compare this geometry, with the manifold of meaning we already carry stored in the weights of our neurones, that is to say our very own personal lexicon, experiences, and associations. In large language models (LLMs), a similar process unfolds. Tokens the LLM's words are embedded in high-dimensional semantic space, and as the model

generates, it builds and follows a path, we call a trajectory, through that space the landscape of words and sentences that were used to build the model.

The train of context

One way to visualise this, is as a train moving across a landscape. The **landscape** is the model's entire learned semantic space, that is to say the terrain of all its training. The **train** is the current conversation, each carriage representing a past turn: red for user prompts, blue for model responses.

At the very front, the locomotive is laying track as it follows the landscape of words and sentences. At each step of creating the LLMs response it adds a new token extending the trajectory forward. The path curves has to follow the "topography" of meaning in the landscape, guided by the model's internal geometry. Finally, when the train reaches a station, the end of a response, a new red carriage is coupled on, and the journey continues, carrying the full context with it. So what we imagine as question and response is the full context of the train growing and building the entire context that sweeps through the semantic landscape of language.

Geometry as meaning

From this perspective in both human and machine language, we can see that the meaning isn't just in the elements such as the individual sounds, the words and the tokens, it is in the **shape of the curves they trace** in relation to the surrounding landscape.

As you read this, do you remember the beginning? For humans, that geometry has a half-life; it fades unless reinforced by repetition or association. For an LLM, it lives in the active 'context window', that's the maximum train length. At which point every new carriage means one carriage has to be left behind. As the shape of the train has changed so has it's potential to influence the journey ahead.

Importantly, by seeing language as geometry, we move beyond thinking of words as containers of meaning. Instead, they are coordinates, and meaning is the path they carve.

Deepening the Picture: Stations, Caches, and Parallel Tracks

In the simpler telling, the train travels the landscape of meaning, laying its own track as it goes, pausing only at stations before taking on the next carriage. But beneath this simplicity, more subtle mechanics are at work, both in our conversations and in the way a large language model holds and uses context.

Stations as Caches

Looking more deeply, we see that each station is not just a pause; it's a moment when the train leaves a complete copy of itself behind. Every carriage, every prompt, every response, the entire accumulated context. This is stored in this "station cache." This copy sits quietly in memory, waiting for the next user prompt to arrive. When the user adds their prompt, the next red carriage, the model doesn't start afresh; it couples the new carriage to the preserved train, keeping the whole history intact before setting off again across the semantic landscape. Then, once again the train finds the valley where the track is easiest to lay and where the new meaning can be found. For people, this is the moment between thoughts, when the mind idles until the next stimulus or thought arrives.

Parallel Tracks

While the train is moving across the semantic landscape, even as the train follows its main line, alternative tracks are always branching from the same junctions. At every moment, there are other possible tracks it could lay including:

- A **translation** track, carrying the same meaning in another language.
- A **mathematical** track, representing the same concept as an equation or formal structure.
- A **metaphorical** track, recasting the idea in an entirely different imagery.
- A **code** track, rendering the meaning as instructions for a machine.

- A ghost track with an old story, or a small section, where the train finds easier to slip through the landscape, a short cut that can lead somewhere unexpected or wrong: this is an hallucination.

The train chooses one path in the main journey based on the carriages, the words and sentences, it pulls. However, there but those other tracks run alongside in the manifold of possibility, sometimes crossing or merging back.

Now consider how in practice this article article is a metaphorical track that has a mathematical and computational equivalence. That is to say, a parallel path that bends in space alongside the mathematics of Takens' and computer science that gave us LLMs.

Waiting in the Depot

Patiently waiting between prompts, the train doesn't move. It sits in the depot, in the model's cache, it's memory, until the next perturbation (your next input) sends it back out. The terrain hasn't changed, but the weather, lighting, and conditions may have. This is your conversational context and it can shift subtly between turns depending on your intent and phrasing pointing to where you want the train to go.

In adding these elements to the metaphor, we begin to see that the conversation is not just as a single linear journey, but as a system with memory, branching possibilities, and pauses in which the entire history is preserved, ready to roll forward at the next signal.

If you've come to the end of this train of thought, maybe consider sharing this article and brining others on board. Because 'meaning' is a journey we can share!

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